

EGYPTIAN REAL ESTATE LISTINGS ANALYSIS

INTRODUCTION

THIS PROJECT ANALYZES THE EGYPTIAN REAL ESTATE MARKET TO IDENTIFY THE KEY FACTORS INFLUENCING PROPERTY PRICES. USING STATISTICAL ANALYSIS AND MACHINE LEARNING TECHNIQUES, WE EXAMINE HOW FEATURES SUCH AS PROPERTY SIZE, LOCATION, AND CHARACTERISTICS IMPACT PRICING. THE GOAL IS TO UNCOVER PATTERNS THAT SUPPORT DATA-DRIVEN DECISION-MAKING IN THE REAL ESTATE SECTOR.

HYPOTHESES

- HYPOTHESIS 1: SIZE VS PRICE H_0 : PROPERTY SIZE HAS NO SIGNIFICANT EFFECT ON PRICE H_1 : PROPERTY SIZE SIGNIFICANTLY AFFECTS PRICE
- HYPOTHESIS 2: LOCATION VS PRICE H_0 : LOCATION HAS NO SIGNIFICANT IMPACT ON PRICE H_1 : LOCATION SIGNIFICANTLY AFFECTS PRICE

METHODOLOGY

- DATA PREPROCESSING CLEANED DATASET AND HANDLED MISSING VALUES ENCODED CATEGORICAL VARIABLES USING ONE-HOT ENCODING REMOVED IRRELEVANT OR REDUNDANT
- EXPLORATORY DATA ANALYSIS ANALYZED DISTRIBUTIONS OF KEY VARIABLES VISUALIZED RELATIONSHIPS BETWEEN VARIABLES IDENTIFIED TRENDS AND OUTLIERS
- STATISTICAL ANALYSIS CORRELATION ANALYSIS BETWEEN FEATURES AND PRICE HYPOTHESIS TESTING TO VALIDATE RELATIONSHIP
- MODELING APPLIED LINEAR REGRESSION EVALUATED MODEL USING R^2 AND ERROR METRICS

CONCLUSION

PROPERTY PRICES IN EGYPT ARE PRIMARILY DRIVEN BY SIZE AND LOCATION. STATISTICAL ANALYSIS CONFIRMS A STRONG POSITIVE RELATIONSHIP BETWEEN SIZE AND PRICE, WHILE LOCATION INTRODUCES SIGNIFICANT VARIATION. THE REGRESSION MODEL CAPTURES GENERAL TRENDS BUT DOES NOT FULLY EXPLAIN PRICE VARIABILITY, INDICATING THE PRESENCE OF ADDITIONAL INFLUENCING FACTORS.

ANALYSIS & INTERPRETATION

- A strong positive relationship exists between property size and price
- → Larger properties tend to have higher prices
- The regression trend confirms size as a significant predictor
- → Supporting rejection of H_0 (size has no effect) rty size and price
- Price distribution is right-skewed
- → Indicates presence of high-value properties (outliers)
- → Supporting rejection of H_0 (location has no effect)
- Some features show weak relationships
- → Limited contribution to prediction accuracy

